**DS230 Final project (Winter2026)**
**Submission deadline: May 19, 2026, at 8:00am**
**Discussions: May 19. Unless otherwise instructed, the deadlines are final!**

Predicting online news popularity and engagement analytics

For this semester's final project, you will work on a realistic and challenging machine learning problem involving online media analytics. Your objective is to predict the popularity and engagement level of online news articles using article metadata, textual characteristics, publishing behavior, sentiment indicators, and content-related features.

You will use the **Online news popularity dataset** from the UCI Machine Learning Repository.

Dataset Link:
https://archive.ics.uci.edu/ml/datasets/Online+News+Popularity

The dataset contains a collection of news articles published by Mashable over approximately two years. Each article includes multiple engineered features that describe content structure, linguistic properties, publication timing, keyword statistics, sentiment measurements, and article metadata.

The project is designed to be **medium-to-tough difficulty (approximately 70% core requirements + 20% advanced analysis)** and follows the same philosophy as previous semesters, emphasizing:

- Heavy preprocessing and feature engineering
- Classification and regression tasks
- Model comparison and tuning
- Explainability and interpretability
- Robustness and stress testing
- Visualization-heavy analysis
- Scientific reasoning and reproducibility

## High-Level Tasks

Your project must include the following major components:

1. Full Exploratory Data Analysis (EDA)
2. Data preprocessing and cleaning
3. Feature engineering and feature selection
4. Two supervised learning tasks:
    - **Task A — Classification:** Predict whether an article will become popular.
    - **Task B — Regression:** Predict the expected number of shares for an article.
5. Implementation of all required machine learning models
6. Hyperparameter tuning and model selection
7. Explainability and interpretability analysis
8. Robustness testing and stress experiments

9. Interactive visualization/dashboard
10. Final report and presentation

## Dataset Understanding

The dataset contains approximately 39,000 observations and over 50 engineered features.

Examples of available variables include:

- Number of words in title/content
- Number of hyperlinks
- Number of images/videos
- Keyword statistics
- Subjectivity and sentiment measures
- Publishing weekday
- Topic category indicators
- Self-reference statistics
- Shares count (target variable)

Students are expected to fully understand all attributes and explain how features may influence article popularity.

## Task A — Classification

Students must formulate a binary classification problem.
Examples include:

- Predict whether an article is **popular vs non-popular**.
- Popularity threshold may be defined as:
  - Above median shares
  - Top 25% most-shared articles
  - Any justified threshold

Students must justify their threshold selection.

**Required Classification Models**
You must implement and compare:

- Logistic Regression (L1/L2 regularization)
- K-Nearest Neighbors Classifier
- Support Vector Machine (linear + kernel)
- Decision Tree Classifier
- Random Forest Classifier
- Gradient Boosting Classifier (XGBoost or LightGBM) — required
- Optional: Stacking Classifier

## Task B — Regression

Students must predict article engagement numerically.

Examples include:

- Predict total number of shares
- Predict log-transformed shares
- Predict popularity score derived from shares

Students must justify target selection.

**Required Regression Models**

You must implement and compare:

- Linear Regression
- Ridge Regression
- Lasso Regression
- Elastic Net
- K-Nearest Neighbors Regressor
- Support Vector Regressor
- Decision Tree Regressor
- Random Forest Regressor
- Gradient Boosting Regressor (XGBoost or LightGBM) — required
- Optional: Stacked Regressor

## Full Preprocessing Study

Your preprocessing must include:

**1. Data Inspection**

- Dataset structure overview
- Data types
- Memory usage analysis
- Duplicate detection
- Missing value inspection

**2. Exploratory Data Analysis**

Required visuals include:

- Histograms
- Density plots
- Boxplots
- Scatter plots
- Pairwise relationships
- Correlation heatmaps
- Feature distributions
- Popularity distribution analysis
- Topic/category analysis
- Day-of-week popularity trends
- Sentiment vs shares analysis

**3. Cleaning & Missing Values**

Students must:
- Detect missing values
- Decide whether to impute/remove
- Justify chosen methods
- Compare different imputation strategies if needed

**4. Outlier Analysis**
Students must:
- Detect outliers
- Compare removal vs winsorization
- Study influence on performance

**5. Feature Scaling**
Students must evaluate:
- StandardScaler
- MinMaxScaler
- RobustScaler

Justify which models require scaling.

**6. Encoding**
Students must:
- Detect categorical variables
- Apply suitable encoding methods
- Compare encoding impact

**7. Feature Engineering (Mandatory)**
Students must create at least several engineered features such as:
- Interaction features
- Ratios
- Keyword density measures
- Sentiment-weighted engagement score
- Publishing-time transformations
- Weekend vs weekday indicators
- Article complexity score
- Log transformations
- Polynomial features

**8. Feature Selection**
Students must compare:
- Correlation-based filtering
- Recursive Feature Elimination
- Tree-based importance selection
- PCA or dimensionality reduction

## Required Evaluation Metrics

**Classification**
Students must compute:
- Accuracy
- Precision
- Recall
- F1-score
- ROC Curve + AUC
- Precision-Recall Curve
- Confusion Matrix
- Calibration Curve
- Brier Score
- MCC (optional but recommended)

**Regression**
Students must compute:
- MAE
- MSE
- RMSE
- $R^2$
- Adjusted $R^2$
- Residual Analysis
- Q-Q Plot
- Heteroscedasticity testing

## Model Comparison Requirements

Students must include:
- Learning curves
- Validation vs training performance
- Bias-variance discussion
- Statistical comparison between models
- At least one statistical significance experiment
    - Paired t-test
    - Wilcoxon signed-rank test

## Explainability Requirements

Students must use explainability tools.

**Required Explainability**
For tree-based models:
- SHAP Summary Plot

- SHAP Dependence Plot
- Feature Importance Ranking

For linear models:

- Coefficient analysis
- Regularization effects

Students must explain:

- Which features matter most
- Why certain features influence popularity
- Interpretability differences across models

**Robustness & Stress Testing**
Students must evaluate model stability.

Required experiments include:
**Noise Injection**
Add Gaussian noise to selected numeric variables.

Evaluate degradation under:

- Low noise
- Medium noise
- High noise

**Feature Removal**
Remove important features and evaluate impact.

**Reduced Training Data**
Train using:

- 10%
- 30%
- 50%
- 100%

Compare learning behavior.

**Outlier Sensitivity**
Inject synthetic outliers and compare performance.

## Visualization Requirements

Students must include a strong visualization component.

Required figures include:

- Histograms
- Density plots

- Correlation heatmaps
- Feature importance plots
- Actual vs predicted plots
- Residual plots
- Confusion matrices
- ROC curves
- PR curves
- Learning curves
- Hyperparameter heatmaps
- SHAP visualizations
- PCA visualization (if used)
- Decision boundary plots for selected models

**Interactive Dashboard Requirement**

Students must create one interactive visualization/dashboard.

Suggested technologies:

- Streamlit
- Plotly Dash
- Voila

Dashboard should allow:

- Model selection
- Metric comparison
- Hyperparameter exploration
- Feature importance display
- Prediction testing

## Cross Validation & Hyperparameter Tuning

Students must include:

- GridSearchCV
- RandomizedSearchCV
- Nested CV OR justified alternative
- Cross-validation discussion
- Final hold-out evaluation

Students must document:

- Parameter grids
- Best parameters
- Search strategy
- Runtime discussion

## Deliverables

### 1. Code Repository

Must include:

- Jupyter notebooks or scripts
- Modular code
- README
- requirements.txt or environment.yml
- Clear folder organization

### 2. Final Report (PDF)

Must include:

- EDA summary
- Preprocessing decisions
- Feature engineering
- Model comparisons
- Metrics tables
- Explainability section
- Robustness results
- Final conclusions
- Recommended model

### 3. Interactive Dashboard

Required.

### 4. Presentation / Discussion

Students must be able to explain:

- All preprocessing decisions
- Model choices
- Evaluation metrics
- Explainability findings
- Weaknesses and limitations

## Suggested Grading Rubric

| Component | Weight |
|---|---|
| Preprocessing + EDA | 20% |
| Modeling + Evaluation | 25% |
| Regression Task | 15% |
| Classification Task | 15% |
| Explainability + Robustness | 15% |
| Code Quality + Reproducibility | 5% |
| Presentation + Understanding | 5% |

## Extra Credit Options

Students may choose up to three:

1. Neural network baseline
2. AutoML comparison
3. Model compression / ONNX export
4. Counterfactual explanations
5. Time-based popularity prediction sequence modeling
6. Topic modeling integration
7. Embedding-based article representations
8. Ensemble stacking
9. Deployment of final model

**Final Advice**

This project is intentionally designed to simulate a real-world data science workflow. The goal is not simply to achieve the best accuracy, but to demonstrate:

- Data understanding
- Scientific reasoning
- Proper preprocessing
- Fair model comparison
- Explainability
- Robustness
- Professional reporting

Focus on producing a polished, reproducible, and scientifically sound project.

**Important notes:**

- For this project, everything you do has to be accompanied with figures, as you learnt in the lectures! Missing figures for any part means that your solution is incomplete!
- We will be checking for similarities between your projects, with the internet, and usage of AI! Write your own code, in your own way!
- Not attending the discussion of the project will result in a *ZERO* no matter how much work you invest in it!
- We will test YOUR UNDERSTANDING! So having all requirements ready, will not necessarily get you a full grade if you can verify your understanding of all required concepts during the discussion!
- Do not change your team without telling us so, during the first week!
- Don't forget the workload distribution of the team members in the readme file!
- Only one submission per team!